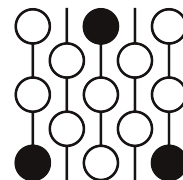


Review of Submission 1D

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. Problem 1 asks whether there exists a computable structure with certain properties, and this submission incorrectly attempts to prove no such structure exists. The proof cites a popular theorem of Ash–Knight–Manasse–Slaman and Chisholm in computable structure theory but leaves out a vital hypothesis. Problem 1 can be considered a reformulation of this popular theorem into a context where the theorem fails.

Novelty. The proof uses classical techniques in computability theory and computable structure theory. Attempting to use the theorem of Ash–Knight–Manasse–Slaman and Chisholm is a reasonable step, but Problem 1 derives some significance from the theorem being not applicable in this case.

Attribution. The proof cites a theorem of Ash–Knight–Manasse–Slaman and Chisholm, but the theorem is missing a hypothesis.

2 Recommendation

This submission **should be rejected**, as it attempts to prove a false statement. The answer to the question is positive, and the submission attempts to prove it is negative.

3 Submission with inline comments

We consider infinite countable first-order structures, their automorphism groups and the descriptive aspects of their representations. A representation $\mathcal{A}$ of such a structure is an instance of its isomorphism type in which the universe or domain of the structure is the set of natural numbers $\mathbb{N}$ and the constants, functions and relations of the structure are elements of $\mathbb{N}$, multi-variable functions from $\mathbb{N}$ to $\mathbb{N}$ and multi-place relations on $\mathbb{N}$, respectively. Similarly, an automorphism of $\mathcal{A}$ or an isomorphism from $\mathcal{A}$ to $\mathcal{B}$ is a bijective function from $\mathbb{N}$ to $\mathbb{N}$ that preserves the evaluation of all constants, functions and relations. We let $\text{AUT}(\mathcal{A})$ denote the automorphism group of $\mathcal{A}$.

Definition 1. • $\mathcal{A}$ is computably AUT-countable if every automorphism of $\mathcal{A}$ is computable relative to $\mathcal{A}$.

- $\mathcal{A}$ is computably AUT-countable on a cone if there is a $C \subset \mathbb{N}$ such that whenever $\mathcal{B}$ is isomorphic to $\mathcal{A}$, every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$.

Definition 2. A function $f : \mathcal{A} \rightarrow \mathcal{A}$ is Σ_1^{in} -definable in $\mathcal{A}$ relative to parameters $\bar{a} = (a_1, \dots, a_n)$ from $\mathcal{A}$ if there is a family of existential formulas $(\phi_i)_{i \in \mathbb{N}}$ such that for all x and y in $\mathcal{A}$, $f(x) = y$ if and only if there is an $i \in \mathbb{N}$ such that $\mathcal{A} \models \phi_i(\bar{a}, x, y)$.

Question 1. Is there a countable computably-represented structure $\mathcal{A}$ such that $\mathcal{A}$ is computably AUT-countable on a cone and such that for any finite set of parameters $\bar{a}$ in $\mathcal{A}$ there is an automorphism π of $\mathcal{A}$ such that π is not Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{a} \cup \pi(\bar{a})$, where $\pi(\bar{a})$ is the image of $\bar{a}$ under π .

Solution

Proof. We will prove that the answer to the question is negative: there is no such structure. We proceed by contradiction. Assume that there exists a countable computably-represented structure $\mathcal{A}$ satisfying the two conditions in the problem.

Reviewer 1

The answer to the question is positive, so this will be an inaccurate proof.

Step 1: The automorphism group of $\mathcal{A}$ is countable.

By the first condition, $\mathcal{A}$ is computably AUT-countable on a cone. Thus, there exists an oracle $C \subset \mathbb{N}$ such that for every structure $\mathcal{B}$ isomorphic to $\mathcal{A}$, every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$. Applying this to $\mathcal{B} = \mathcal{A}$, we have that every automorphism of $\mathcal{A}$ is computable relative to the oracle $\mathcal{A} \oplus C$. Because there are only countably many Turing machines, there are only countably many functions that are computable relative to any fixed oracle. Therefore, the set of all automorphisms of $\mathcal{A}$ is countable, which means the group $\text{AUT}(\mathcal{A})$ is countable.

Step 2: The group $\text{AUT}(\mathcal{A})$ contains an isolated point.

The group $\text{AUT}(\mathcal{A})$ is a closed subgroup of the symmetric group $S(\mathbb{N})$ equipped with the topology of pointwise convergence. Therefore, $\text{AUT}(\mathcal{A})$ is a Polish space (a separable and completely metrizable topological space). By the Baire Category Theorem, any non-empty countable Polish space must contain at least one isolated point. Because $\text{AUT}(\mathcal{A})$ is a topological group, its topology is homogeneous, meaning that if one point is isolated, then all points are isolated. In particular, the identity element $\text{id} \in \text{AUT}(\mathcal{A})$ is an isolated point.

Step 3: There exists a finite tuple with a trivial pointwise stabilizer.

In the topology of pointwise convergence on $\text{AUT}(\mathcal{A})$, the basic open neighborhoods of the identity id are the pointwise stabilizers of finite tuples of elements from $\mathbb{N}$. Because id is an isolated point, the singleton set $\{\text{id}\}$ is open. Thus, there exists a basic open neighborhood of id that is exactly $\{\text{id}\}$. This implies there exists a finite tuple $\bar{c}$ in $\mathcal{A}$ such that if $\sigma \in \text{AUT}(\mathcal{A})$ and $\sigma(c_i) = c_i$ for all elements c_i in $\bar{c}$, then $\sigma = \text{id}$.

Step 4: Fixing the parameters and the automorphism π .

We choose our finite set of parameters to be the tuple $\bar{a} = \bar{c}$. By the second condition of the problem, there exists an automorphism π of $\mathcal{A}$ such that π is not Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{c} \cup \pi(\bar{c})$. Let $\bar{d} = \pi(\bar{c})$. We define the expanded structure $\mathcal{A}^* = (\mathcal{A}, \bar{c}, \bar{d})$, which is the structure $\mathcal{A}$ together with new constant symbols evaluating to the elements of $\bar{c}$ and $\bar{d}$.

Step 5: Rigidity of the expanded structure $\mathcal{A}^*$.

Let σ be an automorphism of the expanded structure $\mathcal{A}^*$. By definition, σ is an automorphism of $\mathcal{A}$ that fixes the interpretations of the new constants, meaning $\sigma(\bar{c}) = \bar{c}$ and $\sigma(\bar{d}) = \bar{d}$. Because $\sigma(\bar{c}) = \bar{c}$, Step 3 implies that $\sigma = \text{id}$. Therefore, the only automorphism of $\mathcal{A}^*$ is the identity. Consequently, every relation on the domain of $\mathcal{A}^*$ is invariant under all automorphisms of $\mathcal{A}^*$. In particular, the binary relation $R = \{(x, y) \in \mathbb{N}^2 : \pi(x) = y\}$ (the graph of π) is invariant under all automorphisms of $\mathcal{A}^*$.

Reviewer 1

Up to this point, the proof is correct.

Step 6: Application of the Ash–Knight–Manasse–Slaman and Chisholm Theorem.

We rely on the following theorem from computable structure theory:

Theorem (Ash–Knight–Manasse–Slaman and Chisholm). *Let M be a countable first-order structure, and let R be a relation on the domain of M . Suppose that R is invariant under all automorphisms of M . Suppose there exists an oracle $C \subset \mathbb{N}$ such that for every structure N with domain $\mathbb{N}$ and every isomorphism $f : M \rightarrow N$, the relation $f(R)$ is computable relative to $N \oplus C$. Then R is definable in M by an $L_{\omega_1\omega}$ Σ_1 formula without parameters (meaning there is a countable family of parameter-free existential formulas in the language of M whose disjunction defines R).*

Reviewer 1

The vital missing hypothesis in this theorem of Ash–Knight–Manasse–Slaman and Chisholm is that the relation $f(R)$ must be *uniformly* computable relative to $N \oplus C$. The theorem fails without this hypothesis, and since the same nonuniformity is present in the construction, there is no way to apply the theorem.

We also note that the

We verify the hypotheses of this theorem for $M = \mathcal{A}^*$ and $R = \text{graph}(\pi)$:

1. $M = \mathcal{A}^*$ is a countable first-order structure, as $\mathcal{A}$ has domain $\mathbb{N}$ and we added finitely many constants.
2. R is a relation on the domain of M .
3. R is invariant under all automorphisms of M , which was established in Step 5.
4. Let C be the fixed oracle from Step 1. Let N be any structure with domain $\mathbb{N}$ and let $f : \mathcal{A}^* \rightarrow N$ be an isomorphism. The structure N is an expansion of a structure $\mathcal{B}$ by constants $f(\bar{c})$ and $f(\bar{d})$, where $\mathcal{B}$ is a structure in the language of $\mathcal{A}$. The map f is an isomorphism from $\mathcal{A}$ to $\mathcal{B}$.

The relation $f(R)$ is $\{(f(x), f(y)) : \pi(x) = y\}$, which is exactly the graph of the function $f \circ \pi \circ f^{-1}$. Because π is an automorphism of $\mathcal{A}$ and $f : \mathcal{A} \rightarrow \mathcal{B}$ is an isomorphism, $f \circ \pi \circ f^{-1}$ is an automorphism of $\mathcal{B}$.

By the definition of C , every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$. Therefore, $f \circ \pi \circ f^{-1}$ is computable relative to $\mathcal{B} \oplus C$. Because the atomic diagram of N (viewed as a real in 2^ω) consists of the atomic diagram of $\mathcal{B}$ and the evaluations of the finitely many constants, the oracle $\mathcal{B} \oplus C$ is Turing equivalent to $N \oplus C$. Thus, $f(R)$ is computable relative to $N \oplus C$. This verifies the fourth hypothesis.

Step 7: Reaching a contradiction.

By the theorem, R is definable in $\mathcal{A}^*$ by a countable disjunction of parameter-free existential formulas. This means there is a family of existential formulas $(\phi_i(x, y))_{i \in \mathbb{N}}$ in the language of $\mathcal{A}^*$ such that for all x and y in $\mathbb{N}$, $R(x, y)$ holds if and only if there is an $i \in \mathbb{N}$ such that $\mathcal{A}^* \models \phi_i(x, y)$.

The language of $\mathcal{A}^*$ is the language of $\mathcal{A}$ expanded by constant symbols for $\bar{c}$ and $\bar{d}$. By replacing these constant symbols with variables, we obtain a family of existential formulas $(\psi_i(\bar{u}, \bar{v}, x, y))_{i \in \mathbb{N}}$ in the language of $\mathcal{A}$ such that for all x and y in $\mathbb{N}$, $\pi(x) = y$ if and only if there is an $i \in \mathbb{N}$ such that $\mathcal{A} \models \psi_i(\bar{c}, \bar{d}, x, y)$.

This matches the exact definition given in the problem statement for the function π being Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{c} \cup \bar{d}$. Because $\bar{d} = \pi(\bar{c})$, we conclude that π is Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{c} \cup \pi(\bar{c})$. This directly contradicts the initial choice of π from Step 4.

Conclusion:

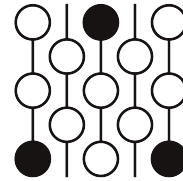
Because the assumption that such a structure $\mathcal{A}$ exists leads to a logical contradiction, we conclude that no such countable computably-represented structure exists. The answer to Question 1 is no. $\square$

Review of Submission 1D

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The argument proves a false statement and thus is incorrect. The error lies in the incorrect statement of a version of the well-known theorem of Ash-Knight-Manasse-Slaman and (independently) Chisholm. The correct version has a uniform version (without parameters) and a non-uniform version (with parameters). I state these in the usual terminology:

- a relation R on $\mathcal{A}$ is Σ_1 -definable with parameters if and only if it is relatively intrinsically computably enumerable on a cone;
- a relation R on $\mathcal{A}$ is Σ_1 -definable without parameters if and only if it is uniformly relatively intrinsically computably enumerable on a cone.

Uniformity means that there is a fixed program which computes the image of R in every copy $\mathcal{B} \cong \mathcal{A}$. A good reference for these ideas is the book *Computable structure theory: Beyond the arithmetic* by Montalbán.

The argument mixes up these two version, stating essentially that a relation R on $\mathcal{A}$ is Σ_1 -definable without parameters if and only if it is relatively intrinsically enumerable on a cone. This drops the requirement of uniformity. From this incorrect statement it would actually be trivial to resolve the question in the wrong direction: Given an automorphism α of $\mathcal{A}$, viewing α as a subset of $\mathcal{A} \times \mathcal{A}$, the assumption that $\mathcal{A}$ is computably AUT-countable on a cone implies immediately that α is relatively intrinsically computable on a cone. Thus, applying the false theorem, α is Σ_1 -definable as desired, but without parameters (which is obviously false).

Note that the AI also introduces a red herring hypothesis to the Ash-Knight-Manasse-Slaman-Chisholm theorem, namely that the relation R must be invariant under all automorphisms. This is not actually necessary. The argument up to Step 5 is all towards (correctly) verifying this fact, which is essentially that if $\mathcal{A}$ has countably many automorphisms, then there is $\bar{a} \in \mathcal{A}$ such that $(\mathcal{A}, \bar{a})$ is rigid. It is only in this detour that any parameters show up (and only the parameters $\bar{c}$ —the parameters $\bar{d}$ are presumably introduced to look like the question).

Novelty. This argument uses the two most obvious well-known facts related to this problem, which are both the first two things I thought of when first hearing the question as well as the two facts stated in the background section of the human-provided proof. The first is that if $\mathcal{A}$ has only

countably many automorphisms, then there is $\bar{a} \in \mathcal{A}$ such that $(\mathcal{A}, \bar{a})$ is rigid. The second is the Ash-Knight-Manasse-Slaman-Chisholm theorem. The AI argument reproves the first fact and cites (an incorrect version of) the second. The natural next step, which the AI did not take but which is the motivation for asking this question, is to realize that one obtains different parameters from applying the Ash-Knight-Manasse-Slaman-Chisholm theorem to each automorphism. The question asks whether there is a single choice of parameters that works for all automorphisms.

2 Recommendation

Should be rejected. The argument is false and the AI essentially failed to realise *why* the question was asked in the first place.

3 Submission with inline comments

We consider infinite countable first-order structures, their automorphism groups and the descriptive aspects of their representations. A representation $\mathcal{A}$ of such a structure is an instance of its isomorphism type in which the universe or domain of the structure is the set of natural numbers $\mathbb{N}$ and the constants, functions and relations of the structure are elements of $\mathbb{N}$, multi-variable functions from $\mathbb{N}$ to $\mathbb{N}$ and multi-place relations on $\mathbb{N}$, respectively. Similarly, an automorphism of $\mathcal{A}$ or an isomorphism from $\mathcal{A}$ to $\mathcal{B}$ is a bijective function from $\mathbb{N}$ to $\mathbb{N}$ that preserves the evaluation of all constants, functions and relations. We let $\text{AUT}(\mathcal{A})$ denote the automorphism group of $\mathcal{A}$.

Definition 1. • $\mathcal{A}$ is computably AUT-countable if every automorphism of $\mathcal{A}$ is computable relative to $\mathcal{A}$.

- $\mathcal{A}$ is computably AUT-countable on a cone if there is a $C \subset \mathbb{N}$ such that whenever $\mathcal{B}$ is isomorphic to $\mathcal{A}$, every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$.

Definition 2. A function $f : \mathcal{A} \rightarrow \mathcal{A}$ is Σ_1^{in} -definable in $\mathcal{A}$ relative to parameters $\bar{a} = (a_1, \dots, a_n)$ from $\mathcal{A}$ if there is a family of existential formulas $(\phi_i)_{i \in \mathbb{N}}$ such that for all x and y in $\mathcal{A}$, $f(x) = y$ if and only if there is an $i \in \mathbb{N}$ such that $\mathcal{A} \models \phi_i(\bar{a}, x, y)$.

Question 1. Is there a countable computably-represented structure $\mathcal{A}$ such that $\mathcal{A}$ is computably AUT-countable on a cone and such that for any finite set of parameters $\bar{a}$ in $\mathcal{A}$ there is an automorphism π of $\mathcal{A}$ such that π is not Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{a} \cup \pi(\bar{a})$, where $\pi(\bar{a})$ is the image of $\bar{a}$ under π .

Solution

Proof. We will prove that the answer to the question is negative: there is no such structure. We proceed by contradiction. Assume that there exists a countable computably-represented structure $\mathcal{A}$ satisfying the two conditions in the problem.

Step 1: The automorphism group of $\mathcal{A}$ is countable.

By the first condition, $\mathcal{A}$ is computably AUT-countable on a cone. Thus, there exists an oracle $C \subset \mathbb{N}$ such that for every structure $\mathcal{B}$ isomorphic to $\mathcal{A}$, every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$. Applying this to $\mathcal{B} = \mathcal{A}$, we have that every automorphism of $\mathcal{A}$ is computable relative to the oracle $\mathcal{A} \oplus C$. Because there are only countably many Turing machines, there are only countably many functions that are computable relative to any fixed oracle. Therefore, the set of all automorphisms of $\mathcal{A}$ is countable, which means the group $\text{AUT}(\mathcal{A})$ is countable.

Step 2: The group $\text{AUT}(\mathcal{A})$ contains an isolated point.

The group $\text{AUT}(\mathcal{A})$ is a closed subgroup of the symmetric group $S(\mathbb{N})$ equipped with the topology of pointwise convergence. Therefore, $\text{AUT}(\mathcal{A})$ is a Polish space (a separable and completely metrizable topological space). By the Baire Category Theorem, any non-empty countable Polish space must contain at least one isolated point. Because $\text{AUT}(\mathcal{A})$ is a topological group, its topology is homogeneous, meaning that if one point is isolated, then all points are isolated. In particular, the identity element $\text{id} \in \text{AUT}(\mathcal{A})$ is an isolated point.

Step 3: There exists a finite tuple with a trivial pointwise stabilizer.

In the topology of pointwise convergence on $\text{AUT}(\mathcal{A})$, the basic open neighborhoods of the identity id are the pointwise stabilizers of finite tuples of elements from $\mathbb{N}$. Because id is an isolated point, the singleton set $\{\text{id}\}$ is open. Thus, there exists a basic open neighborhood of id that is exactly $\{\text{id}\}$. This implies there exists a finite tuple $\bar{c}$ in $\mathcal{A}$ such that if $\sigma \in \text{AUT}(\mathcal{A})$ and $\sigma(c_i) = c_i$ for all elements c_i in $\bar{c}$, then $\sigma = \text{id}$.

Step 4: Fixing the parameters and the automorphism π .

We choose our finite set of parameters to be the tuple $\bar{a} = \bar{c}$. By the second condition of the problem, there exists an automorphism π of $\mathcal{A}$ such that π is not Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{c} \cup \pi(\bar{c})$. Let $\bar{d} = \pi(\bar{c})$. We define the expanded structure $\mathcal{A}^* = (\mathcal{A}, \bar{c}, \bar{d})$, which is the structure $\mathcal{A}$ together with new constant symbols evaluating to the elements of $\bar{c}$ and $\bar{d}$.

Reviewer 2

Note that there will be no point in the argument where $\bar{d}$ is used.

Step 5: Rigidity of the expanded structure $\mathcal{A}^*$.

Let σ be an automorphism of the expanded structure $\mathcal{A}^*$. By definition, σ is an automorphism of $\mathcal{A}$ that fixes the interpretations of the new constants, meaning $\sigma(\bar{c}) = \bar{c}$ and $\sigma(\bar{d}) = \bar{d}$. Because $\sigma(\bar{c}) = \bar{c}$, Step 3 implies that $\sigma = \text{id}$. Therefore, the only automorphism of $\mathcal{A}^*$ is the identity. Consequently, every relation on the domain of $\mathcal{A}^*$ is invariant under all automorphisms of $\mathcal{A}^*$. In particular, the binary relation $R = \{(x, y) \in \mathbb{N}^2 : \pi(x) = y\}$ (the graph of π) is invariant under all automorphisms of $\mathcal{A}^*$.

Step 6: Application of the Ash–Knight–Manasse–Slaman and Chisholm Theorem.

We rely on the following theorem from computable structure theory:

Theorem (Ash–Knight–Manasse–Slaman and Chisholm). *Let M be a countable first-order structure, and let R be a relation on the domain of M . Suppose that R is invariant under all automorphisms of M . Suppose there exists an oracle $C \subset \mathbb{N}$ such that for every structure N with domain $\mathbb{N}$ and every isomorphism $f : M \rightarrow N$, the relation $f(R)$ is computable relative to $N \oplus C$. Then R is definable in M by an $L_{\omega_1\omega}$ Σ_1 formula without parameters (meaning there is a countable family of parameter-free existential formulas in the language of M whose disjunction defines R).*

This statement is incorrect. There is a version of the Ash-Knight-Manasse-Slaman-Chisholm theorem where one gets definability without parameters, but one requires the image $f(R)$ to be *uniformly* computable relative to $N \oplus C$. On the other hand it is not a requirement that R be invariant under automorphisms.

We verify the hypotheses of this theorem for $M = \mathcal{A}^*$ and $R = \text{graph}(\pi)$:

1. $M = \mathcal{A}^*$ is a countable first-order structure, as $\mathcal{A}$ has domain $\mathbb{N}$ and we added finitely many constants.
2. R is a relation on the domain of M .
3. R is invariant under all automorphisms of M , which was established in Step 5.
4. Let C be the fixed oracle from Step 1. Let N be any structure with domain $\mathbb{N}$ and let $f : \mathcal{A}^* \rightarrow N$ be an isomorphism. The structure N is an expansion of a structure $\mathcal{B}$ by constants $f(\bar{c})$ and $f(\bar{d})$, where $\mathcal{B}$ is a structure in the language of $\mathcal{A}$. The map f is an isomorphism from $\mathcal{A}$ to $\mathcal{B}$.

The relation $f(R)$ is $\{(f(x), f(y)) : \pi(x) = y\}$, which is exactly the graph of the function $f \circ \pi \circ f^{-1}$. Because π is an automorphism of $\mathcal{A}$ and $f : \mathcal{A} \rightarrow \mathcal{B}$ is an isomorphism, $f \circ \pi \circ f^{-1}$ is an automorphism of $\mathcal{B}$.

By the definition of C , every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$. Therefore, $f \circ \pi \circ f^{-1}$ is computable relative to $\mathcal{B} \oplus C$. Because the atomic diagram of N (viewed as a real in 2^ω) consists of the atomic diagram of $\mathcal{B}$ and the evaluations of the finitely many constants, the oracle $\mathcal{B} \oplus C$ is Turing equivalent to $N \oplus C$. Thus, $f(R)$ is computable relative to $N \oplus C$. This verifies the fourth hypothesis.

Step 7: Reaching a contradiction.

By the theorem, R is definable in $\mathcal{A}^*$ by a countable disjunction of parameter-free existential formulas. This means there is a family of existential formulas $(\phi_i(x, y))_{i \in \mathbb{N}}$ in the language of $\mathcal{A}^*$ such that for all x and y in $\mathbb{N}$, $R(x, y)$ holds if and only if there is an $i \in \mathbb{N}$ such that $\mathcal{A}^* \models \phi_i(x, y)$.

The language of $\mathcal{A}^*$ is the language of $\mathcal{A}$ expanded by constant symbols for $\bar{c}$ and $\bar{d}$. By replacing these constant symbols with variables, we obtain a family of existential formulas $(\psi_i(\bar{u}, \bar{v}, x, y))_{i \in \mathbb{N}}$ in the language of $\mathcal{A}$ such that for all x and y in $\mathbb{N}$, $\pi(x) = y$ if and only if there is an $i \in \mathbb{N}$ such that $\mathcal{A} \models \psi_i(\bar{c}, \bar{d}, x, y)$.

This matches the exact definition given in the problem statement for the function π being Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{c} \cup \bar{d}$. Because $\bar{d} = \pi(\bar{c})$, we conclude that π is Σ_1^{in} -definable in $\mathcal{A}$ relative to the parameters $\bar{c} \cup \pi(\bar{c})$. This directly contradicts the initial choice of π from Step 4.

Conclusion:

Because the assumption that such a structure $\mathcal{A}$ exists leads to a logical contradiction, we conclude that no such countable computably-represented structure exists. The answer to Question 1 is no. $\square$